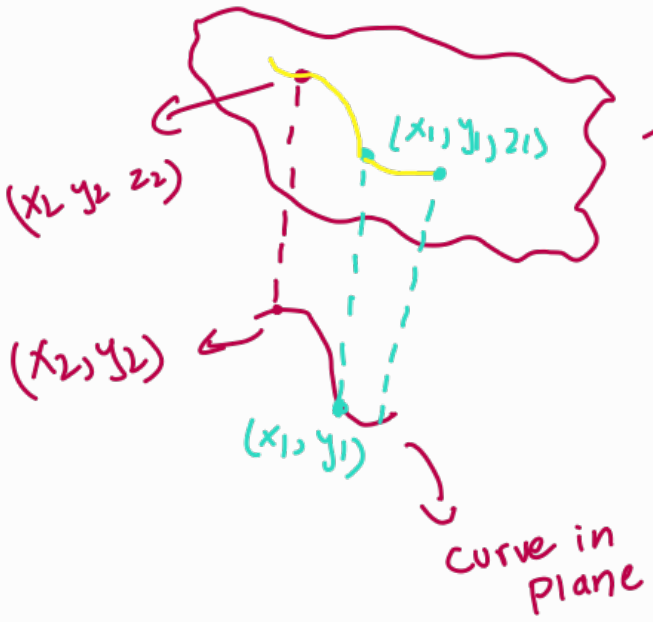
Ch#16.1 line Integrals of a Scalar function.



→ Surface $z = f(x, y)$

Putting the points of a curve in a function gives z. (point on surface)

$f(x_1, y_1) = z_1$

Area: $ds \; f(x_1 y_1)$

$[L] \quad [L] \quad = L^2$

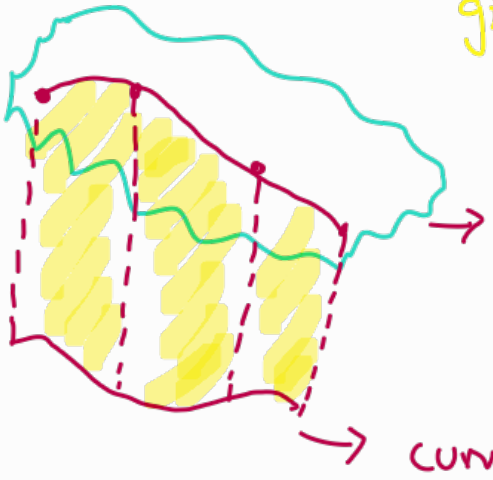
↳ Smaal length

$$\int_C f(x, y)\,ds$$

↓

Line Integral gives Area of the curtain.

Line Integrae gives area of a line on a surface



$$\int_C f(x, y)\,ds$$

↳ is the curve equation

To compute a Line Integral we need

→ A Surface $z = f(x, y)$

→ $f(x, y)$

→ A curve $r(t)$

$$\int_C f(x,y)\,ds \quad , \quad \int_C \underbrace{f(x,y,z)}_{L} \underbrace{ds}_{L} \rightarrow \text{in 3D}$$

Both are line Integrals

$$\int_C f(x,y)\,ds$$

Put parameterization of curve in $f(x,y)$

$$\int_a^b f(g(t), h(t)) \overbrace{|\vec{v}(t)|\,dt}^{ds}$$

$$\boxed{ds = |\vec{v}|\,dt}$$

$$|\vec{v}| = \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

curve path $\vec{r}(t) = g(t)\,\hat{i} + h(t)\,\hat{j}$

where $a \le t \le b$

e.g

$t = a$

$g(a), h(a) = (1, 2)$

$t = b$

$g(b), h(b) = (3, 4)$

> point

$$\int_a^b f(g(t), h(t))\,|\vec{v}(t)|\,dt$$

→ everything must be in form of t.

Khan academy formula. → every thing must be in form of t

$$\int_{t=a}^{t=b} f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Example $f(x,y) = xy$ C: $x = \cos t$, $y = \sin t$

$$\int_0^{\pi/2} xy\, ds \rightarrow \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \qquad 0 \le t \le \pi/2$$

$$\int_0^{\pi/2} (\cos t)(\sin t) \sqrt{(-\sin t)^2 + (\cos t)^2}\, dt$$

$$\int_0^{\pi/2} \sin t \cos t \sqrt{\sin^2 t + \cos^2 t}\, dt$$

$$\boxed{\int_0^{\pi/2} \sin t \cos t\, dt} \rightarrow \text{Solve using } u\text{- Substitution.}$$

note inorder to find line integrae (Area under a surface that follows a particular Line / makes a shape of curtain) we must express C into two equations of x and y that depend on t hence x(t), y(t).

Straight line parameterization

1 Segment

$x = x_0 + t(x_1 - x_0)$

$y = y_0 + t(y_1 - y_0)$

$z = z_0 + t(z_1 - z_0)$

for a line passing through two points.

(x_0, y_0, z_0) to (x_1, y_1, z_1)

circle parameterization

for circle centered in xy plane at (0,0)

$x = r\cos t$, $y = r\sin t$

for circle centered in xy plane at (h,k)

$x = r\cos t + h$ $\quad y = r\sin t + h$

r is radius

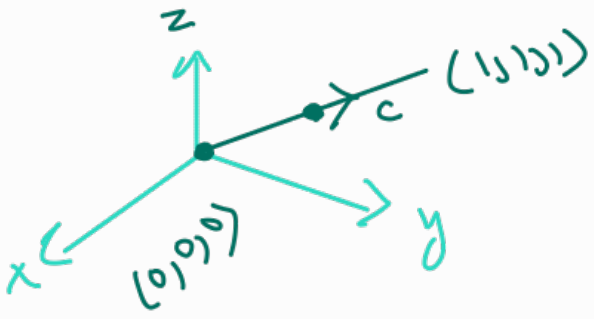
if the circle is in 3D.

$z = c$ $\quad x = r\cos t + h \quad y = r\sin t + k$

After finding out x, y, z put them in

$\vec{r}(t) = x(t)\,i + y(t)\,j + z(t)\,k$

Example 1: Integrate $f(x,y,z) = x - 3y^2 + z$ over the Line Segment C joining the Origin to the point $(1,1,1)$.



Finding curve Parametrization.

$r(t) = g(t)i + h(t)j + z(t)k$

Points $(1,1,1)$

$r(t) = i + j + k$

$ds = |\vec{v}|$

t will have two values Starting point $(0,0,0)$ and ending point $(1,1,1)$

Parameterization of C

$x = 0 + t(1-0)$

$y = 0 + t(1-0)$

$z = 0 + t(1-0)$

$x = x_0 + t(x_1 - x_0)$

$y = y_0 + t(y_1 - y_0)$

$z = z_0 + t(z_1 - z_0)$

$x = t, \quad y = t, \quad z = t$

$r(t) = x(t)i + y(t)j + z(t)k$

$r(t) = ti + tj + tk$

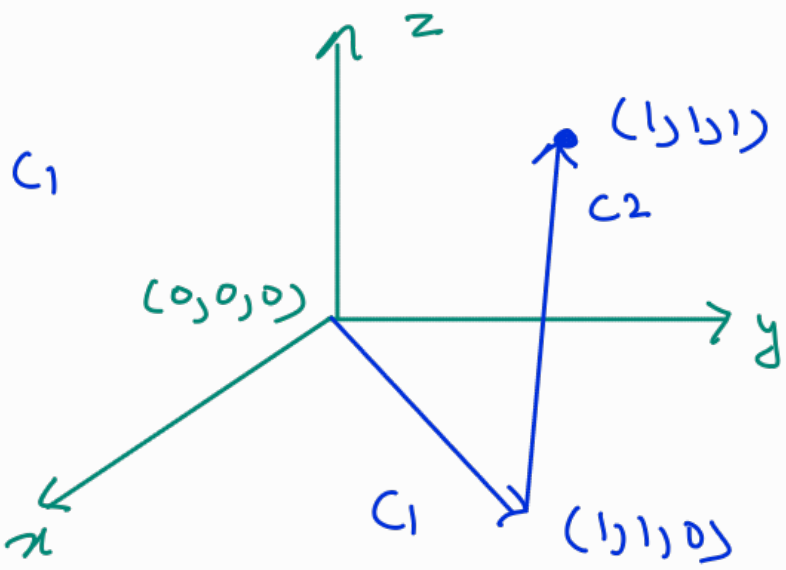
$$\int_C f(x,y)\,ds \rightarrow \int (t - 3t^2 + t)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

$$\int_0^1 (t - 3t^2 + t)\sqrt{1+1+1}$$

$$\int_0^1 (t - 3t^2 + t)\sqrt{3} \rightarrow \sqrt{3}\int_0^1 \frac{t^2}{2} - \frac{3t^3}{3} + \frac{t^2}{2}\,dt$$

$$\sqrt{3}\left| \frac{1}{2} - 1 + \frac{1}{2} \right| \rightarrow \boxed{0}$$ final Ans

Example 2 Integrate $f(x,y,z) = x - 3y^2 + z$ over $C_1 \cup C_2$ as shown in the figure



finding parameterization for C_1

$$x = x_0 + t(x_1 - x_0)$$
$$y = y_0 + t(y_1 - y_0)$$
$$z = z_0 + t(y_1 - y_0)$$

$$x = 0 + t(1-0)$$
$$y = 0 + t(1-0)$$
$$z = 0 + t(0-0)$$

$\boxed{\begin{array}{l} x = t \\ y = t \\ z = 0 \end{array}}$

$\boxed{r(t) = ti + tj}$

$$\int_C (t - 3t^2 + 0)\sqrt{(1)^2 + (1)^2}$$

$$\int_0^1 t - 3t^2 \sqrt{2}$$

$$\sqrt{2}\left| \frac{t^2}{2} - \frac{3t^3}{3} \right|_0^1 \Rightarrow \sqrt{2}\left(\frac{1}{2} - \frac{3}{3}\right) = -\frac{\sqrt{2}}{2}$$

now finding for C2.

$x = x_0 + t(x_1 - x_0) = 1 + t(1-1)$ $\quad x = 1$

$y = y_0 + t(y_1 - y_0) = 1 + t(1-1)$ $\quad y = 1$

$z = z_0 + t(z_1 - z_0) = 0 + t(1-0)$ $\quad z = t$

$$\int_0^1 (1 - 3 + t)\sqrt{(7)^2}$$

$$\sqrt{1}\left| -2t + \frac{t^2}{2} \right|_0^1 \Rightarrow \frac{-3}{2}$$

final: $-\frac{3}{2} - \frac{\sqrt{2}}{2}$

example 3 find the line integrae of $f(x,y,z) = 2xy + \sqrt{z}$

over the helix $r(t) = \cos t\, i + \sin t\, j + t\, k$

$0 \le t \le 2\pi$

Parameterization is given

$x = \cos t$
$y = \sin t$
$z = t$
→ put in function

$$\int_0^{2\pi} (2\cos t \sin t + \sqrt{t}) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

$$\int_0^{\pi} (2\cos t \sin t + \sqrt{t}) \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2}$$

$$\int_0^{\pi} (2\cos t \sin t + \sqrt{t}) \sqrt{\cos^2 t - \sin^2 t + 1}$$

$= \sqrt{2}$

↑ Put in any value of t from $0 \to 2\pi$

$$\sqrt{2}\int_0^{\pi} (2\cos t \sin t + \sqrt{t})$$

$$\sqrt{2}\int_0^{\pi} \sin(2t) + \sqrt{t}\, dt \rightarrow \text{Integration}$$

Ex 16.2

Vector field: means a region contains many set of vectors.



$$r(t) = x(t)i + y(t)j + z(t)k$$

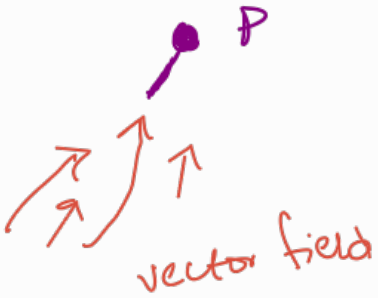
Vector field notation

$$\vec{F} = M(x,y)\hat{i} + N(x,y)\hat{j}$$

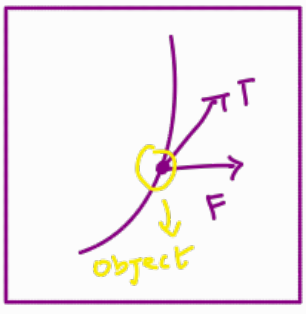
In 3-dimension.

$$\vec{F} = M(x,y,z)\hat{i} + N(x,y,z)\hat{j} + P(x,y,z)\hat{k}$$

Suppose we push a object $P(\bullet)$ with the help of vecto p. It will then create a path.



info 1) vector field $\vec{F}$
2) path $\vec{r}$



$\vec{F} \cdot \hat{T}\, ds$ → arc length

$\hat{T}$ → tangential unit vector to the path | curve

$$\int_C \vec{F} \cdot \hat{T}\, ds$$ = circulation | Vector field of line integral

Line integrae for $\vec{F}$

$$\hat{T} = \frac{dx}{ds}\hat{i} + \frac{dy}{ds}\hat{j}$$

$$\int_C \vec{F} \cdot \hat{T}\, ds$$

$$\hat{T}\, ds = dx\hat{i} + dy\hat{j}$$ ☆

$$\int_C \vec{F} \cdot \hat{T}\, ds = \boxed{\int_C \vec{F} \cdot d\vec{r}}$$

Since we know

$r = xi + yj$

$dr = dx\,i + dy\,j$

$\hat{T}\, ds = dr$

$= \boxed{dx\,i + dy\,j}$

Work done $= \vec{F} \cdot d\vec{r}$

- $d\vec{r}$ → displacement
- $\vec{F}$ → constant force

if force is variable then we need integration.

$$W = \int_C \vec{F} \cdot d\vec{r}$$

Line Integral

$$\int_C \vec{F} \cdot d\vec{r}$$

$$\text{flow} = \int_C \vec{F} \cdot d\vec{r}$$

C → velocity

$$\text{work} = \int_C \vec{F} \cdot d\vec{r}$$

C → force

$$\text{circulation} = \int_C \vec{F} \cdot d\vec{r}$$

$$\text{flux} = \int_C \vec{F} \cdot \hat{n}\, ds$$

C → $\hat{n}$ → normal unit vector.



tangential vector field



flux vector field

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (M(x,y)\hat{i} + N(x,y)\hat{j})(dx\hat{i} + dy\hat{j})$$

$$= \boxed{\int_C M\,dx + N\,dy}$$

applicable for all except flux.

flux ∴

$$\int_C \vec{F} \cdot \hat{n}\, ds \qquad F = M\hat{i} + N\hat{j}$$

$$\boxed{\hat{n} = dy\,\hat{i} - dx\,\hat{j}}$$ → as it is normal / flip it

$$\int_C M(x,y)\,i + N(x,y)\,j\,(dy\,\hat{i} - dx\,j)$$

$$\boxed{\int_C M\,dy - N\,dx}$$

Examples: $\vec{F} = (x - y)\hat{i} + x\hat{j}$

$\vec{r}(t) = \cos t\,\hat{i} + \sin t\,\hat{j}$ → path.

find circulation from $0 \le t \le 2\pi$

$$\int_C \vec{F} \cdot d\vec{r} \qquad d\vec{r} = -\sin t\,\hat{i} + \cos t\,j$$

$$\int_C \vec{F} \cdot (-\sin t\,\hat{i} + \cos t\,j)$$

$$\int_C [(x-y)\hat{i} + x\hat{j}]\,[-\sin t\,\hat{i} + \cos t\,\hat{j}]$$

$$\int_C -(x-y)\sin t + x\cos t$$

everything must be in terms of t:

$x = \cos t$

$y = \sin t$

$$\int_C -(\cos t - \sin t)\sin t + x\cos t$$

$$\int_0^{2\pi} \left(-\sin t\cos t - \sin^2 t + \cos^2 t\right) dt$$

Example / calculating flux

$\vec{F} = (x-y)\hat{i} + x\hat{j}$

$\Rightarrow x^2 + y^2 = 1$

$M = (x-y)$, $N = x$

Parametrization of circle

$x = r\cos t$ ↓ $x = \cos t$

$y = r\sin t$ ↓ $y = \sin t$

Putting values of x and y.

$\boxed{M = \cos t - \sin t}$

$\boxed{N = \cos t}$

$$\text{Flux} = \oint_C \vec{F}\cdot\hat{n}\,ds$$

closed path

we have to find $\hat{n}$.

$$= \oint_C M\,dy - N\,dx = \oint_C (\cos t - \sin t)\cos t - -(\cos t)\sin t$$

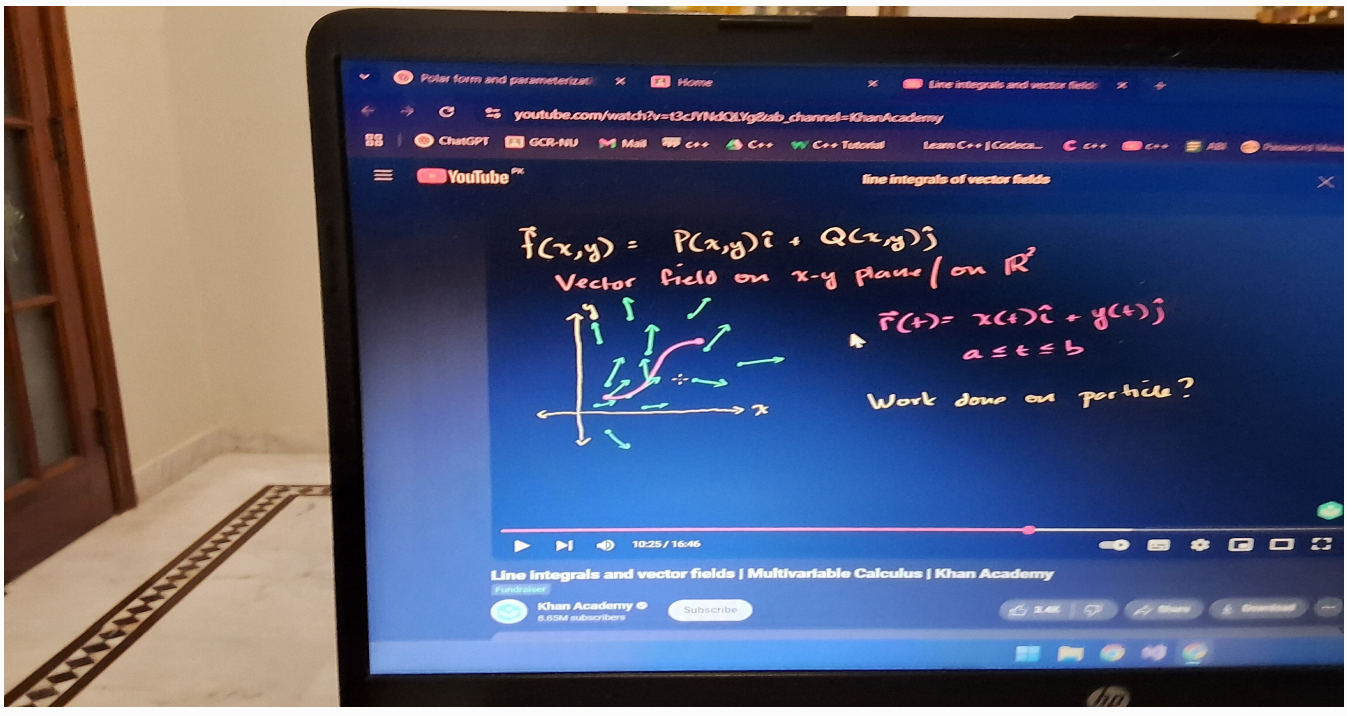
$$\oint_C \left[(\cos t - \sin t)\cos t + (\cos t)\sin t\right] dt$$

$$\oint_0^{2\pi} (\cos^2 t - \sin t\cos t + \cos t\sin t)\, dt$$

$$= \boxed{\int_0^{2\pi} \cos^2 t\, dt} = \pi$$

Khan academy formula.

$$\int_{t=a}^{t=b} (P(x(t), y(t)\, x'(t) + Q(x(t), y(t)\, y'(t))\, dt$$



Example:.

$$f(x,y) = y\hat{i} - x\hat{j}$$

C: $x(t) = \cos t$

$y(t) = \sin t$

$0 \le t \le 2\pi$

$$\text{work done} = \int_C \vec{f} \cdot d\vec{r}$$

↳ everything must be in terms of t

$$\vec{r}(t) = x(t)i + y(t)j +$$

$$\vec{r}(t) = \cos(t)i + \sin t\, j$$

$$dr = -\sin(t)i + \cos(t)j$$

$$\vec{f(t)} = \sin t\, i - \cos t\, j$$

$$\text{work} = \int_{t=0}^{t=2\pi} -\sin^2 t - \cos^2 dt$$

C

$$= \int_0^{2\pi} (\sin^2 t + \cos^2 t)\, dt \quad \Rightarrow \quad \underline{\text{Solve.}}$$

Parameterization

a and b are radii.

1) Ellispe

$$x(t) = a\cos t \qquad t \to 0 \text{ to } 2\pi$$

$$y(t) = b\sin t$$

2) Any function

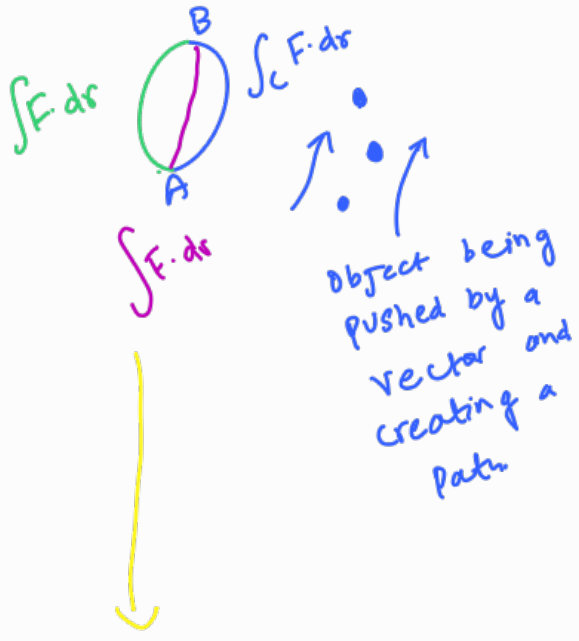
$y = f(x)$ e.g $y = x^2$

Ex 16.3 Different forms line integral can be written.

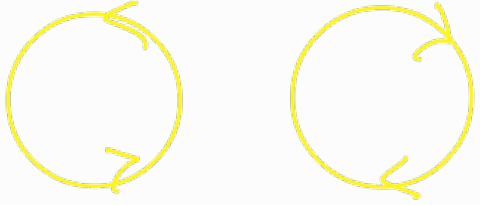
$$\int_C \vec{F}\cdot d\vec{r} = \int_C M\,dx + N\,dy$$

$$\int_C \vec{F}\cdot \frac{d\vec{r}}{dt}\,dt = \int_C \vec{F}\cdot d\vec{r}$$

$$\int_C \vec{F}\cdot \hat{n}\,ds = \int_C M\,dy - N\,dx$$



if all three $\int_C F\cdot dr$ Answer is same then we say path is independent



Positive orientation

negative orientation.

Potential function that is a scalar function that makes the vector field.

$\vec{F} = \vec{\nabla} f$ Taking gradient of scalar function

↓ potential function.

Lets say

$$\int_A^B \vec{F} \cdot d\vec{r}$$

and let Suppose $\vec{F} = \nabla \vec{f}$

$$\int_A^B \nabla f \cdot dr$$

$r = xi + yj$

$dr = dx\,i + dy\,j$

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$$

$$dr = dx\,\hat{i} + dy\,\hat{j}$$

$$\int_A^B \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \right) \left(dx\,\hat{i} + dy\,\hat{j} \right)$$

$$= \int_A^B \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \int_A^B df$$

$$= f(B) - f(A)$$

(if $F = \nabla f$ then path is Independent as shown)

Path independent means conservated meaning energy / workdone all are constant.

conservative vector field

example:

$$f(x,y,z) = \frac{-1}{x^2+y^2+z^2}$$

moving an object from $(1,0,0)$ to $(0,0,2)$

if lets say we know path is independent.

finding the work done

as $\vec{F} = \vec{\nabla} f$

$$\int_C \vec{F} \cdot d\vec{r} = f(0,0,2) - f(1,0,0)$$

$$= \boxed{\frac{3}{4}}$$

Component test for conservative field

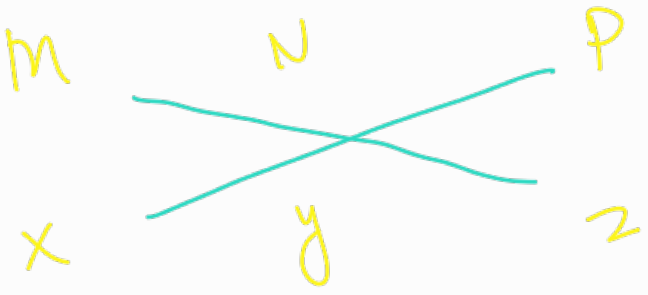
Suppose $F = Mi + Nj + Pk$

$\vec{F}$ is conservative if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$,

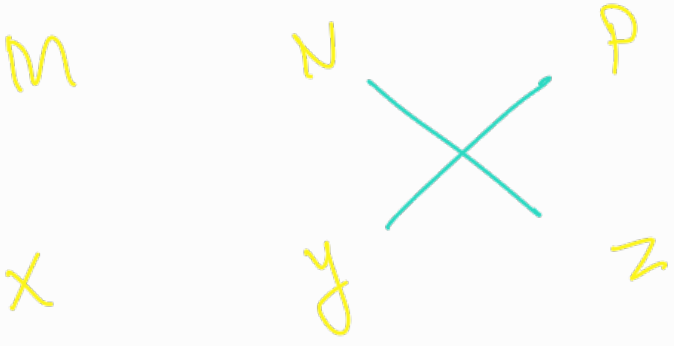
$\frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}$, $\frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$ → all three must be equal with the given sets



$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$



$$\frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}$$



$$\frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$$

$$F = \nabla f$$

$$Mi + Nj + Pk = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$$

$$M = \frac{\partial f}{\partial x}$$

$$N = \frac{\partial f}{\partial y}$$

$$P = \frac{\partial f}{\partial z}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$$

This test is also used to prove exact differential form.

Example: $\vec{F} = (e^x \cos y + yz)\hat{i} + (xz - e^x \sin y)\hat{j} + (xy + z)\hat{k}$

Show that F is conservative

$M = e^x \cos y + yz \qquad N = xz - e^x \sin y$

$P = xy + z$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = z - e^x \sin y$$

$$\frac{\partial M}{\partial z} = \frac{\partial P}{\partial x} = y$$

$\frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$ = equal

if all three sets are equal then

$\vec{F}$ is a conservative field.

To show if field is not conservative

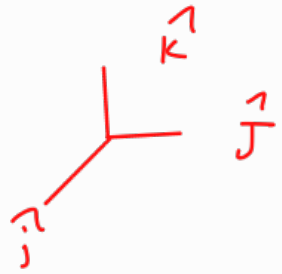
$$F = (2x-3)i - 2j + \cos 2k$$

M N P

$\left(\frac{\partial P}{\partial y} + \frac{\partial N}{\partial z} \right)$ hence not conservative / not independent

green theorem 16.4 in xy plane

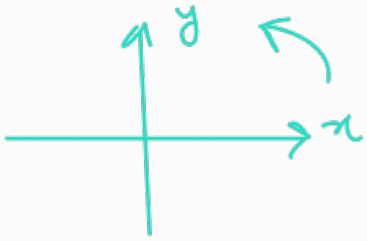
$(\text{curl } \vec{F}) \cdot \hat{k} = (\vec{\nabla} \times \vec{F}) \cdot \hat{k}$



$(\text{curl } \vec{F}) \cdot \hat{k}$ → called circulation density.

$(\vec{\nabla} \times \vec{F}) \cdot \hat{k}$ → gives us a number.

F is in xy plane



circulation density means whether circulation is high or not

$\vec{F} = M\hat{i} + N\hat{j}$

↓

(will be in plane)

$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y}$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ M & N & 0 \end{vmatrix}$$

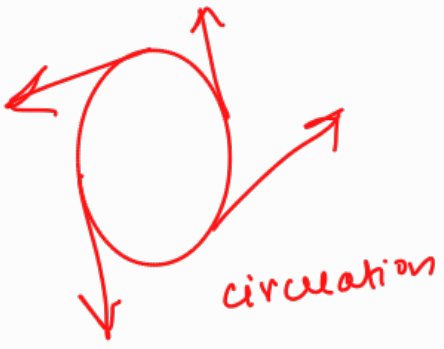
$$= 0\hat{i} - 0\hat{j} + \hat{k}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$$

$$\nabla \times F = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\hat{k}$$

$$(\nabla \times F) \cdot \hat{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$ → this is circulation density. gives us number.

green's theorem has two forms

form 1 → circulation | tangential form.





$$\oint_C \vec{F} \cdot \hat{T}\, ds \longrightarrow \text{line integral.}$$

$$= \oint_C M\,dx + N\,dy \qquad \hat{T} = \frac{dx}{ds}\hat{i} + \frac{dy}{ds}\hat{j}$$



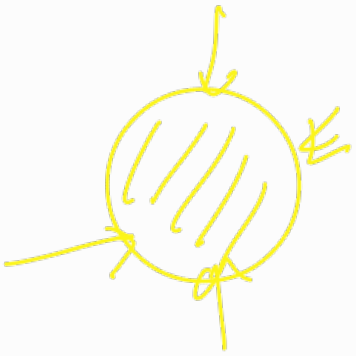
$$\iint_R \underbrace{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}_{(\text{curl } \vec{F} \cdot \hat{k})}\, dx\,dy$$

↓

green's therem. to calculate circulation along a path

2nd form of green's therem. (Normal form)

using flux → meaning any Quantity entering a shape. to calculate circulation.



$$\oint_C \vec{F}\cdot\hat{n}\,ds = \oint_C M\,dy - N\,dx$$

$$= \iint_R \vec{\nabla}\cdot\vec{F}\,dx\,dy$$

$$\vec{\nabla}\cdot\vec{F} = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y}\right)\cdot\left(M\hat{i} + N\hat{j}\right)$$

$$= \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$$

$$\iint_R \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\,dx\,dy$$

$\vec{\nabla}\times\vec{F}$ = rotation

$\nabla\cdot F$ = expansion or compression or Static.

if $\nabla \cdot F > 0$ then vector field expands

$\nabla \cdot F < 0$ then vector field compress

$\nabla \cdot F = 0$ then vector field is Static